

Attention to pain localization and unpleasantness discriminates the functions of the medial and lateral pain systems

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Abstract

Functional imaging studies have identified a matrix of structures in the brain that respond to noxious stimuli. Within this matrix, a division of function between sensory-discriminative and affective responses has so far been demonstrated by manipulating either pain intensity or unpleasantness under hypnosis in two different normal volunteer groups studied on separate occasions. Our study used positron emission tomography (PET) to demonstrate this division of function under more natural conditions in a healthy group of volunteers, using a CO₂ laser to provide nociceptive stimuli that selectively activate A-delta and C-fibres without contamination by touch sensations. We measured the differential cerebral responses to noxious and innocuous laser stimuli during conditions of selective attention to either the unpleasantness or location of the stimuli. Attention to location increased responses in the contralateral (right) primary somatosensory and inferior parietal cortices. This result implies that these components of the lateral pain system are concerned mainly with the localization of pain. In contrast, attention to unpleasantness increased responses in bilateral perigenual cingulate and orbitofrontal cortices, contralateral (right) amygdala, ipsilateral (left) hypothalamus, posterior insula, M1 and frontal pole. These areas comprise key components of the medial pain and neuroendocrine systems and the results suggest that they have a role in the affective response to pain. Our results indicate the importance of attentional effects on the pattern of nociceptive processing in the brain. They also provide the first clear demonstration, within a single experiment, of a major division of function within the neural pain matrix.

Introduction

It is now been widely accepted that pain is a multidimensional experience and has sensory-discriminative, affective, motivational and evaluative components. These are likely to be processed within a neural matrix (Melzack *et al.*, 1968). Based on post mortem reports and neurosurgical observations, a division of function between the lateral and medial components of the human pain processing system of the brain was proposed several decades ago (Bowsher, 1957; Albe-Fessard *et al.*, 1985). Functional imaging studies have enabled researchers to investigate this proposal further. The lateral pain system comprises the lateral thalamic nuclei and the primary and secondary somatosensory cortices (SI and SII, respectively). It is somatotopic and was initially thought to subserve the sensory-discriminative aspects of pain such as localization, intensity and duration. The insular cortex also has somatotopic nociceptive inputs and may be involved in integrating these with inputs from other sensory modalities (Ostrowsky *et al.*, 2002). The medial pain system has been considered to be slow and its thalamo-cortical projections are polysynaptic, relative to the lateral system (fewer synapses from lateral thalamus to SI and SII compared to medial thalamic projections to cortical areas),

nonsomatotopic and to process the affective components of pain (Treede *et al.*, 1999). It includes the medial thalamic nuclei, anterior cingulate (ACC) and prefrontal cortices and structures concerned with the processing of fear, such as the amygdala. This is supported by clinical observations of the alleviation of affective components of pain by selective anterior cingulate lesions (Foltz & White, 1962) and effects of lesions in the region of SI on the sensory-discriminative components of pain (Ploner *et al.*, 1999).

Functional imaging studies have attempted to demonstrate this division of function within the pain matrix by attentional manipulations and hypnosis (Rainville *et al.*, 1997; Peyron *et al.*, 1999; Tölle *et al.*, 1999; Hofbauer *et al.*, 2001). These studies have used intensity of pain to access the sensory-discriminative components of pain and unpleasantness to access the affective components. It is now evident that intensity is encoded throughout the pain matrix (Coghill *et al.*, 1999; Derbyshire *et al.*, 1997), although discrete intensity-associated areas have been identified within the mid-cingulate cortex (Büchel *et al.*, 2002) and SI (Hofbauer *et al.*, 2001). Therefore, it seems that intensity coding is not a specific function of the lateral pain system. In addition, pain unpleasantness and intensity are highly correlated (Rainville *et al.*, 1999) and therefore difficult to dissociate without resorting to techniques such as hypnosis. For these reasons we selected localization rather than intensity as a purer index of sensory-discriminative processing for this study.

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Our study compared brain activations during the localization and unpleasantness rating of identical noxious and innocuous heat stimuli. This is the first functional imaging study to compare sensory and affective nociceptive processing in the same group of subjects using attention to 'unpleasantness' and 'localization' of nociceptive stimuli. The results of this study might have therapeutic implications and lead to future pharmacological and psychophysiological studies in developing new treatments for pain.

Materials and methods

Experiments conformed to The Code of Ethics of the World Medical Association (Declaration of Helsinki) (Rickham, 1964). Approval was obtained from the relevant ethics committees (The Salford and Trafford Local Research Ethics Committee and the National Hospital for Neurology and Neurosurgery (UCLH NHS Trust) and Institute of Neurology (UCL) joint research ethics committee). Seventeen right-handed, healthy male volunteers (mean age 28 years; range 26–33) gave their written informed consent to participate in the study. All were free of medical and psychiatric symptoms and ongoing pain.

Stimulus

A CO₂ laser stimulator (Derbyshire *et al.*, 1997) was used to provide well-controlled and spatially defined phasic radiant heat stimuli that selectively activate A-delta and C-fibre nociceptive afferents, sparing the touch-sensitive A-beta fibres (Bromm *et al.*, 1984). Subjects received either noxious or innocuous laser stimuli during the performance of three different attentional tasks (Table 1).

PET procedure

Brain activity was measured with ¹⁵O-labelled water and PET. Each subject underwent a single training session in the 6 weeks before positron emission tomography (PET) scanning. During this training session the right dorsal forearm was stimulated. On the day of PET scanning the left dorsal forearm was used. Four stimulation points were marked on the forearm, the distance between which (7 cm vertically and 3 cm horizontally) was determined so that subjects were able to discriminate areas of laser stimulation with an accuracy of at least 75% correct responses per stimulation block. Four ramps of increasing laser energy were delivered to the marked sites in order to obtain individual pain ratings on a 0–10 numerical rating scale (NRS: 0 = no pain; 4 = just painful; 7 = moderate pain; 10 = unbearable pain). Laser energies consistently rated as level 7 (painful) and level 3 (nonpainful) were individually noted and used to train subjects on the different attentional tasks. On the day of scanning the same laser energies were used as a starting point for re-determining individuals' level 3 and level 7 energies.

During PET scanning subjects performed three attentional tasks (attend to stimulus unpleasantness, attend to stimulus location and

control) twice at each stimulus intensity (noxious and innocuous) in separate blocks, thereby giving 12 stimulation blocks in total and hence 12 scans. Laser stimuli were pseudo-randomly moved between the four stimulation points during all tasks. Subjects were blind to stimulus location. Written task instructions and rating scales were displayed on a monitor within a subject's field of view. During the 'attention to unpleasantness' task subjects were instructed to selectively attend to how unpleasant they found the laser pulses and rate each stimulus on a 0–4 affective rating scale, where 0 = no sensation felt, 1 = pleasant warmth, 2 = neither pleasant nor unpleasant, 3 = moderately unpleasant, and 4 = strongly unpleasant. Subjects responded to a cue on the monitor by pressing the appropriate number on a response pad held in the right hand. In the 'attention to location' task subjects were instructed to selectively concentrate on the location of stimuli and to indicate at which of the four sites they were felt by pressing the appropriate button on a response pad when cued to do so. If no stimulus was felt they chose 0. In the control task they were asked when cued to respond to successive laser stimuli by pressing a button on a response pad in sequence without attending to any particular aspect of the stimuli. After each block of stimulation subjects gave an average pain rating for that block using the NRS and also reported the approximate percentage of stimuli perceived as painful. They also reported how confidently they performed the required task ('totally confident', 'very confident', 'slightly confident', and 'not at all confident'). The motor components of all three tasks were identical, as were the number of response options, and subjects could choose '0' if a stimulus was not felt. Subjects were asked specifically not to rate the intensity of each separate stimulus, as this would cause a division of attention between unpleasantness and intensity and location and intensity in the 'attention to unpleasantness' and 'attention to location' tasks, respectively.

Our design conformed to a 2 × 3 factorial design with two levels of stimuli (noxious vs. innocuous) and three levels of attentional set (unpleasantness, location and control; Table 1).

Image acquisition

Before PET scanning, magnetic resonance imaging (MRI) scans (160 contiguous 1-mm-thick slices) were acquired on a Siemens 2.0 T VISION system to acquire axial T1 weighted structural images for anatomical coregistration. PET data (63 slices) were acquired with an ECAT HR + scanning system (CTI Siemens, Knoxville, TN, USA) in three-dimensional mode with septa retracted. Correction for attenuation was made with a transmission scan recorded using a 68 Ge/Ga external ring source before each session. Images were reconstructed by filtered-back projection to give a resolution of ~6 mm at full width, half maximum and displayed in a 128 × 128 pixel format with 43 planes, rendering the voxels approximately cubic. For each scan, 330 MBq H₂¹⁵O were flushed through a venous cannula in the right antecubital vein with normal saline over 20 s at a rate of 10 mL/min by an automatic pump. After a delay of about 35 s, a rise in head counts was detected that peaked about 30–40 s later. Data were acquired in 90-s frames, beginning 5 s before the rising phase of the head count curve. A total of 12 scans were performed at intervals of 8 min. The delivery of laser stimuli to the forearm began 30 s before the start of a scan and stopped after 90 s. Each scan encompassed a block of 14–16 stimuli depending on the delay in the radioactivity reaching brain.

Data analysis

This study intended to compare attentional modulation of pain-related responses in the brain or, in other words, to compare cortical blood

TABLE 1. Factorial analysis (2 × 3 design)

	Noxious	Innocuous
Attend to Unpleasantness	1	2
Attend to Location	3	4
Control	5	6

Interaction: increases under attention to unpleasantness = (1 – 2) – (3 – 4); increases under attention to location = (3 – 4) – (1 – 2) = –1 × increases under attention to unpleasantness.

flow responses to different attentional conditions during noxious and innocuous stimulation. SPM99b (<http://www.fil.ion.ucl.ac.uk/spm>) was used for all analyses. To correct for head movement between scans all images were realigned to the first one to create a mean PET image. Each subject's MRI scan was then coregistered to the mean PET image. Each realigned set of images from every subject was then reoriented into a standardized stereotaxic anatomical space of the average magnetic resonance image provided by the Montreal Neurological Institute. To increase the signal-to-noise ratio and to accommodate variability in functional anatomy, each image was smoothed with a Gaussian filter of 16 mm. A correction was made for global changes in regional cerebral blood flow (rCBF) between scans. Differences between one condition and another were assessed in the appropriate contrast using the *t*-statistics. This analysis was performed for each pixel and the resulting set of *t*-values constituted a statistical parametric map (SPM {*t*}). Statistical comparisons of blood flow between conditions were performed using random effects analysis. This corrects for variability between subjects so that a single subject does not drive the results. A small volume correction was applied using regional masks to calculate corrected statistics ($P < 0.01$) across the whole brain in all analyses. (Worsley *et al.*, 1996).

In the next step, two separate correlation analyses across the group were performed using the affective (not unpleasant for innocuous stimuli) ratings. The first one correlated the affective ratings with rCBF in the scans when subjects received only noxious stimuli. In the second one, affective ratings were correlated with rCBF in the scans during innocuous stimulation. Significant activations are reported following a small volume correction applied using regional masks to calculate corrected statistics ($P < 0.01$) across the whole brain.

Results

Behavioural results

The affective (unpleasantness) ratings during noxious stimulation were significantly higher (3.4 ± 0.5 , corresponding to moderate unpleasantness on the 0–4 unpleasantness scale) than those during innocuous stimulation (1.8 ± 0.5 ; $P < 0.01$, paired *t*-test; Fig. 1). The mean

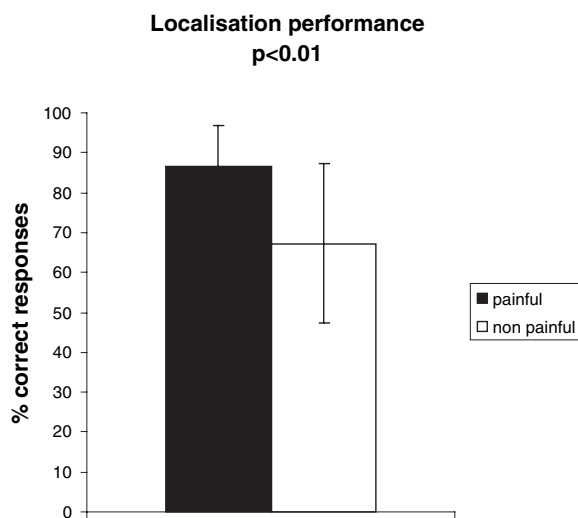


FIG. 1. Mean unpleasantness ratings during the 'attention to unpleasantness task' and subjective performance in 'attention to localization' task during noxious and innocuous stimulation. Error bars indicate SEM ($P < 0.01$, paired *t*-test). (Unpleasantness rating scale: 0, no sensation felt; 1, pleasant warmth; 2, neither pleasant nor unpleasant; 3, moderately unpleasant; 4, strongly unpleasant).

percentage of correct responses in the 'attention to location' task was $86.7 \pm 9.8\%$ and $67.2 \pm 18.2\%$ ($P < 0.01$, paired *t*-test) during noxious and innocuous stimulation, respectively (Fig. 1). The mean retrospective pain rating for all painful conditions was 4.9 ± 0.7 (painful on the NRS) and for nonpainful conditions was 2.4 ± 0.6 (nonpainful on the NRS; $P < 0.01$, paired *t*-test; note that subjects gave an average retrospective pain rating using NRS after each scan for all the tasks). The retrospective percentage painful responses were $77.3 \pm 10\%$ for the unpleasantness task, $75.8 \pm 5\%$ localization task and $77.9 \pm 8\%$ for the control task. There was no significant difference in the retrospective responses among the three tasks ($P > 0.1$, paired *t*-test). In a post-task questionnaire majority of the subjects confirmed that they were all 'very confident' in attending to the stimulus attribute of interest. For each level of stimulation (noxious or innocuous), the perceived pain intensity (retrospective pain ratings) was matched across all three tasks, therefore any differences between the tasks were due to attentional set.

rCBF responses

An initial analysis included a categorical comparison of the main effect of pain in all conditions collapsed together and one of the effect of pain in scans without attentional instructions. The main effect of pain [(1 + 3 + 5) vs. (2 + 4 + 6); Table 1] was associated with increased relative rCBF (activation) in ipsilateral (left) thalamus, bilateral anterior cingulate cortices (ACC) (BA 24), ipsilateral (left) anterior insula, contralateral (right) posterior insula, bilateral basal ganglia and SII, contralateral (right) SI, bilateral precuneus, supplementary motor area, cerebellum and upper brainstem up to the pons (corrected $P < 0.05$). The effect of pain without attentional instructions [(5 vs. 6); Table 1] showed activation in the bilateral anterior cingulate cortices (BA 24), posterior insula/SII, basal ganglia and cerebellum (corrected $P < 0.05$). However, activations in thalamus and SI did not reach statistical significance.

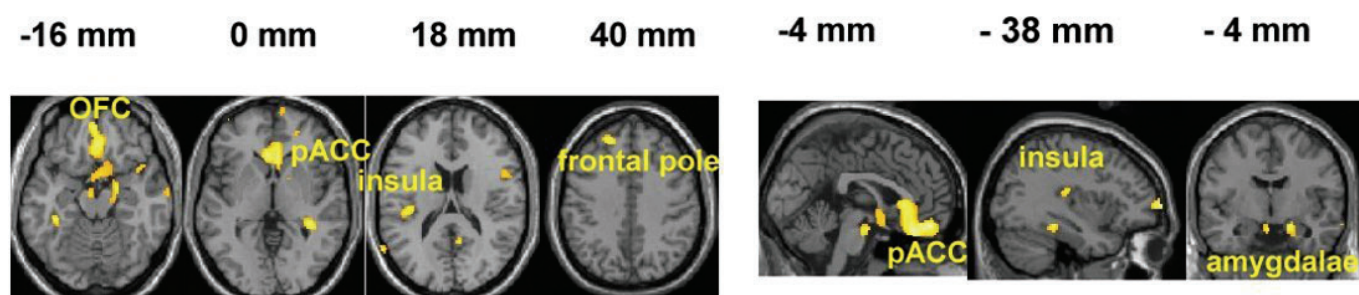
Factorial analysis

This analysis enabled us identify the brain areas that would be more (or less) activated by painful stimulation during attention to unpleasantness than during attention to localization and vice versa, i.e. this is an attentional context-by-stimulation interaction. We tested separately for relative increases in pain-related responses caused by attention to unpleasantness and attention to location. The innocuous condition was always compared to the noxious one and the ensuing pain-related response compared under different levels of attention (see contrasts in Table 1). There was an augmentation of pain-related activation bilaterally in the perigenual ACC, right amygdala, and left hypothalamus when comparing 'attention to unpleasantness' with 'attention to localization'. In addition, there was significantly greater bilateral activation of the orbitofrontal cortex, left posterior insula, MI, and frontal pole. There was increased pain-related activation in the right SI and inferior parietal cortex with 'attention to localization' relative to 'attention to unpleasantness' (Table 2; Fig. 2) and activations in the left precuneus, left MI and right cerebellum failed to reach statistical significance. All results are reported at $P < 0.05$ after small volume corrections using regional masks (Friston *et al.*, 1995).

Correlation analysis

In this analysis, subjective affective ratings for noxious and innocuous stimuli were correlated with rCBF in the corresponding scans (Table 3; Fig. 3). Statistical parametric maps (SPMs) created by this analysis indicate activations that correlate with the affective dimension of

Attention to unpleasantness



Attention to location

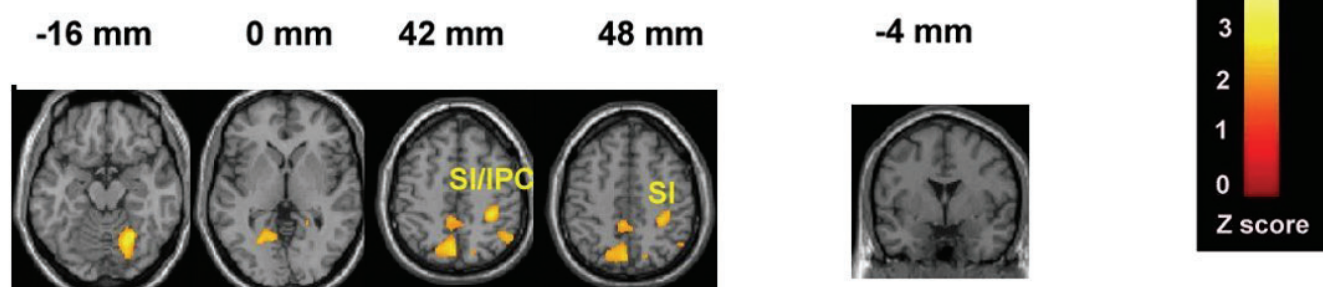


FIG. 2.

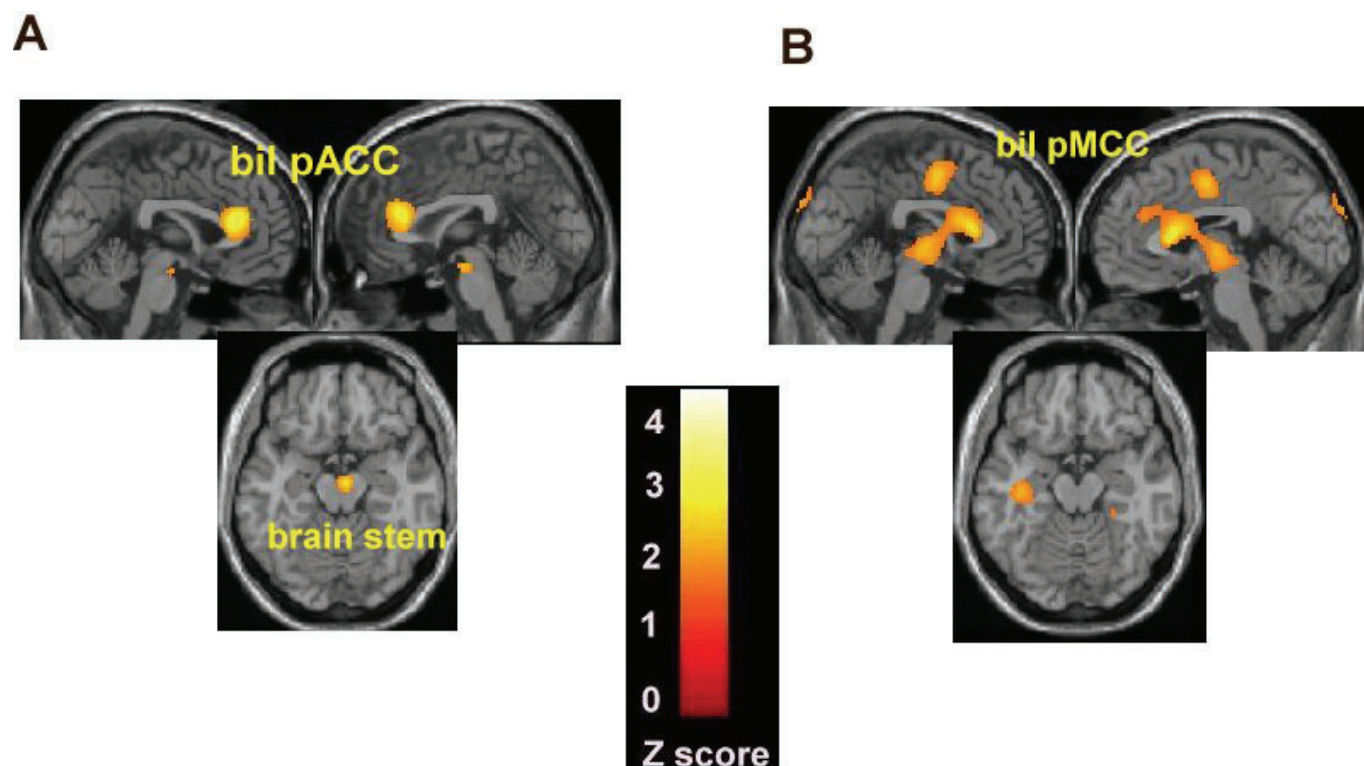


FIG. 3.

TABLE 2. The areas of increased activation during 'attention to unpleasantness' and 'attention to localization' conditions

Brain region	MNI coordinates				<i>T</i> -score	<i>Z</i> -score	<i>K</i> _E	<i>P</i> -values (SV corrected)
	<i>x</i>	<i>y</i>	<i>z</i>	BA				
Interaction between pain and unpleasantness								
(R) pACC	6	32	−8	32	4.83	3.74	343	0.01
(L) pACC	−4	34	2	32	4.18	3.38	343	0.00
(R) OFC	4	38	−22	11	4.45	3.54	623	0.03
(L) OFC	−4	50	−20	11	4.45	3.54	623	0.03
(L) frontal pole	−32	68	6	10	4.25	3.54	94	0.03
(R) amygdala	20	−4	−21	—	4.09	3.34	215	0.03
(L) hypothalamus	−4	−2	−8	—	2.90	2.56	55	0.05
(L) MI	−55	−8	45	4 & 6	4.20	3.40	94	0.00
(L) posterior insula	−42	−26	16	—	3.53	2.99	55	0.01
Interaction between pain and localization								
(R) SI	39	−30	51	3,1,2	4.25	3.54	95	0.05
(L) inferior parietal cortex	46	−54	58	40	5.53	4.45	115	0.007

The *P*-values (SV corrected) indicate *P*-values after small volume correction; (R), right; (L), left; pACC, perigenual anterior cingulate cortex; OFC, orbitofrontal cortex; MI, primary motor cortex; SI, primary somatosensory cortex; BA, Brodmann's area; K_E, cluster size; MNI, Montreal Neurological Institute.

TABLE 3. The areas significantly correlated with subjective affective ratings of stimuli during the 'attention to unpleasantness' tasks

Brain region	MNI coordinates				<i>T</i> -score	Z-score	<i>K</i> _E	<i>P</i> -values (SV corrected)
	<i>x</i>	<i>y</i>	<i>z</i>	BA				
Correlation analysis with affective (unpleasantness) ratings (noxious stimuli)								
(R) aMCC	4	30	20	32/24	4.75	3.68	253	0.00
(L) aMCC	−4	30	20	32/24	4.45	3.54	253	0.00
(L) MI	−52	10	45	4 & 6	5.99	4.68	215	0.00
Upper brainstem	6	−12	−20	–	3.35	3.35	55	0.00
Correlation analysis with affective (unpleasantness) ratings (innocuous stimuli)								
(R) pMCC	4	−6	42	24	3.32	2.75	253	0.00
(L) pMCC	−6	−4	38	24	3.32	2.75	253	0.00
(R) MI	60	−4	46	4 & 6	3.04	2.60	155	0.00

The *P*-values (SV corrected) indicate *P*-values corrected for small volume centred on the coordinates shown along the *x*, *y* and *z* axes. MI, primary motor cortex; SI, primary somatosensory cortex; ACC, anterior cingulate cortex; BA, Brodmann's area; K_E, cluster size; MNI, Montreal Neurological Institute.

noxious and innocuous heat. The affective ratings during noxious stimulation correlated with activations in bilateral ACC (BA 32/24) including perigenual ACC and anterior mid cingulate cortex (aMCC), left-sided MI and upper brainstem. The affective ratings during innocuous stimulation correlated with pMCC bilaterally and right sided MI. Activations in the cingulate cortex were thresholded at a corrected *P*-value <0.05 following small volume correction. Correlations between thalamic rCBF and the affective ratings in the scans taken during innocuous stimulations failed to reach statistical significance.

Discussion

The present study has identified a clear division of function in the main components of the cerebral nociceptive system. This has been demonstrated using completely matched noxious and innocuous stimuli in a single cohort of volunteers when the only difference in conditions was the attentional instruction. SI and inferior parietal cortices were activated more during attention to localization, whereas perigenual ACC, orbitofrontal cortex (OFC), frontal pole, amygdala,

FIG. 2. PET images showing areas of altered cerebral blood flow when attending to the unpleasantness vs. the localization of an experimental heat stimulus. Attention to unpleasantness was associated with an increased cerebral blood flow in bilateral orbitofrontal (OFC) and perigenual cingulate cortices (pACC), contralateral (right) amygdala, ipsilateral (left) hypothalamus, posterior insula, MI and frontal pole. Attention to localization was associated with an increased cerebral blood flow in contralateral (right) primary somatosensory cortex (SI) and inferior parietal cortex (IPC). The MNI (Montreal Neurological Institute) coordinates for the axial sections are indicated in mm on *z* axis, sagittal section at *z* = -4 mm and the coronal sections are at *y* = -4 mm. The correlates are represented as statistical maps thresholded at corrected *P* < 0.05 following small volume correction.

FIG. 3. Correlation analysis between perceived affective (unpleasantness) ratings and cerebral blood flow during noxious and innocuous stimulation. (A) Positive correlation between blood flow during noxious stimulation and subjective ratings of unpleasantness. The correlation was maximal within bilateral perigenual cingulate cortices (bil pACC) and right upper brainstem (contralateral to the side of stimulation). (B) Positive correlation between blood flow during innocuous stimulation and subjective affective ratings (unpleasantness). This was maximal in bilateral midcingulate cortices (MCC). Note that during each PET scan subjects received either noxious or innocuous stimuli and gave an affective (unpleasantness) rating after each stimulus. The MNI (Montreal Neurological Institute) coordinates of the illustrated slices are *z* = -16 mm for the axial sections and *y* = ± 4 mm for the sagittal sections. The correlates are represented as statistical maps thresholded at corrected *P* < 0.05 following small volume correction.

posterior insula, MI and hypothalamus were more activated during attention to unpleasantness. These results suggest a broad division of function between components of the lateral and medial pain systems, confirming the primary hypothesis. As far as the authors are aware, this is the first functional imaging study to demonstrate such a clear distinction between affective and sensory-discriminative pain processing during attention to 'unpleasantness' and 'localization' of the nociceptive stimuli without recourse to hypnosis.

The results of the main effect of pain and the effect of pain without attentional instructions are consistent with most of the previous reports of pain experiments (see reviews by Peyron *et al.*, 1999; Derbyshire, 2000). However, in comparison with the previous pain studies, our results failed to show any significant contralateral activations of the thalamus in the main effect of pain. Ipsilateral thalamic activation has been reported in a few pain studies (Casey *et al.*, 2001; Adler *et al.*, 1997; Derbyshire *et al.*, 1998). These variations could be attributed to differences in the experimental paradigms and smoothing across the midline.

In this paper the discussion will be focused on the effects of attention to the affective and localization aspects of pain, which was the main aim of this experiment.

Cingulate cortex

The ACC has been divided into perigenual ACC and mid-cingulate cortex (MCC). The MCC has recently been further divided into posterior (pMCC) and anterior (aMCC) divisions (Vogt *et al.*, 2003a). The aMCC is thought to be more concerned with fear and affective processing and is anatomically and functionally distinct from the posterior division of the MCC (pMCC) that is more concerned with executive functions. The unique anatomy of the ACC, with much of the circuitry converging on large layer V efferent projection neurons, might suggest that its computational function is related to matching appropriate motor outputs to afferent inputs (Devinsky *et al.*, 1995).

Evidence for the involvement of the ACC in processing the affective components of pain comes from both clinical observations (Foltz *et al.*, 1962) and PET studies (Derbyshire *et al.*, 1994; Tölle *et al.*, 1999). A recent fMRI study by Singer *et al.* (2004) suggests a role of the ACC in affect. In this study, healthy volunteers experienced painful stimulus and compared it to that experienced by their loved ones present in the same room. ACC was activated in common for 'self' and 'other' conditions and empathy scores positively correlated with ACC responses. It also suggests that pain and empathy share the same brain region involved in affective processing. The factorial analysis in our study demonstrates significantly greater pain-specific activity in the perigenual ACC (BA 32) when attending to the unpleasantness of pain compared with attending to its localization. In addition, the correlation analysis of rCBF and affective ratings of noxious stimuli revealed an area of perigenual ACC extending into the aMCC recently described by Vogt *et al.* (2003a). These observations provide robust evidence for the role of these structures in processing the affective components of pain.

The perigenual ACC in particular appears to be concerned with affective responses (George *et al.*, 1996) including vocalization, autonomic control and fear (Bancaud & Talairach, 1992; Bush *et al.*, 2000). The perigenual ACC seems to be activated in studies where pain is strongly aversive, such as those using tonic cold pain (Kwan *et al.*, 2000) or capsaicin-treated skin (Andersson *et al.*, 1997; Lorenz *et al.*, 2002). This could explain why the perigenual ACC is less commonly activated in studies using experimental pain, except when

the pain stimulus is very unpleasant (allodynia, clinical pain) or attention is directed to the unpleasantness of pain, as in this study.

Our results are quite distinct from those of previous studies in which pain unpleasantness and intensity were manipulated using hypnosis (Rainville *et al.*, 1997). These showed a correlation between MCC cortical response and unpleasantness. The area of MCC activated in this study was slightly posterior (pMCC as per Vogt *et al.*, 2003a) to the one in our study. A PET study by Faymonville *et al.* (2000) differed to the study by Rainville *et al.* (1997) in that hypnosis was induced when subjects experienced pain without any suggestions to alter perceived pain intensity and unpleasantness. The results of this study showed that both sensory and affective responses to noxious stimulation reduced during hypnosis and this reduction is mediated by the MCC (same region as in the study by Rainville *et al.*). This suggests that the MCC possibly has a role in both affective and sensory processing of pain. Our study allows a clear distinction between sensory-discriminative (in this case localization) and affective components of a nociceptive stimulus (as discussed earlier) in a single study, without recourse to hypnosis. Hypnosis might result in a level of conflict between instructions and perceptions that results in confounding activations. There is increasing evidence that the pMCC is mainly concerned with executive functions such as response selection (Frith, 2001), monitoring, and error detection (Devinsky *et al.*, 1995). However, a review of functional imaging results and a greater connectivity with the amygdala suggest that anterior MCC is more concerned with fear (Vogt & Vogt, 2003b) and hence may be involved in processing the affective aspect of pain.

The role of the cingulate cortex in processing a variety of cognitive, motor and nociceptive information is well described (Vogt *et al.*, 1993; Devinsky *et al.*, 1995; Paus *et al.*, 1998; Bush *et al.*, 2000). The observation that the anterior cingulate cortex is the most commonly activated structure in functional imaging studies of pain (Derbyshire, 2000) suggests that it has a pivotal role in nociceptive processing. There is substantial variability of the location of anterior cingulate responses to nociceptive stimuli (Derbyshire, 2000). These locations extend from the perigenual ACC (Vogt *et al.*, 1996; Lorenz *et al.*, 2002), also observed in our study, to the more posterior areas of the MCC (Bentley *et al.*, 2003; Vogt *et al.*, 1996). This finding raises the issue of both pain-specific and more generalized divisions of cognitive function within cingulate cortex.

The MCC is also an important component of the anterior attentional system (Posner & Petersen, 1990; Bush *et al.*, 2000), although nociceptive responses appear to locate to discrete areas that are distinct from those concerned with these more general attentional functions (Derbyshire *et al.*, 1998). Understanding the dynamic interactions between these areas of ACC will provide a better understanding of their relative contribution to the human pain experience.

Our results also demonstrate that cingulate responses are exquisitely sensitive to attentional instructions, as has been previously suggested (Peyron *et al.*, 1999) and illustrate the importance of such cortically driven 'top-down' effects (Frith, 2001). This is consistent with observations of other 'top-down' effects on the pain matrix during anticipation (Porro *et al.*, 2002) and distraction (Bantick *et al.*, 2002; Valet *et al.*, 2004). The fMRI studies by Bantick *et al.* (2002) and Valet *et al.* (2004), using a cognitive distraction task (stroop) during nociceptive stimulation demonstrated increases in orbitofrontal and pACC responses indicating a role of pACC in cognitive modulation of pain. However, in a study by Seminowicz *et al.* (2004), pain modulation of emotional stroop was associated with attenuation in the posterior cingulate cortex. Our results also fit in with the large-scale neurocognitive network model of Mesulam *et al.* (see Mesulam, 1990; Morecraft *et al.*, 1993), which proposed cingulate cortex as the main

contributor to a motivational map that interacts with a perceptual map provided by posterior parietal cortex. These results and ideas also support the potential use of psychological approaches in pain therapy.

The activation of perigenual ACC during attention to unpleasantness is consistent with its relatively high concentration of opioid receptors (Vogt *et al.*, 1995; Jones *et al.*, 1999) compared to executive areas of cingulate cortex and with perigenual changes in occupation by endogenous opioid peptides during chronic pain (Jones *et al.*, 1999; Spetea *et al.*, 2002). There are relatively high concentrations of opioid receptors in the higher association cortices and components of the limbic system, including the amygdala, and low concentrations in primary sensory, motor and visual cortices. This finding has led to a proposal that they may have some function related to the modulation of selective attention in addition to their more direct role in modulation of nociception (Lewis *et al.*, 1981). This notion may be relevant to recent observations of common ACC responses in placebo and opiate analgesia (Petrovic *et al.*, 2002).

Orbitofrontal cortex, amygdala, insula, M1 and hypothalamus

During 'attention to unpleasantness', the activation of the orbitofrontal cortex (OFC), amygdala and hypothalamus is interesting in the context of the role of these structures in emotional processing, reward, aversive conditioning, autonomic regulation and fear (Rolls, 1999). Medial thalamic nuclei, basolateral amygdala and the ACC participate in a circuit involved in avoidance learning (Maren *et al.*, 1991; Rolls, 1999). The OFC is known to have anatomical connections with areas of the limbic system and hence may have a role in modulating affective processes and autonomic regulation. These areas have reciprocal anatomical connections with each other (Aggleton *et al.*, 1980; Carmichael & Price, 1995). Activation of the OFC has been consistently observed in functional imaging studies of clinical (Rosen *et al.*, 1994; Hsieh *et al.*, 1995, 1996a) and experimental pain (Bantick *et al.*, 2002; Tölle *et al.*, 1999; Lorenz *et al.*, 2002; Petrovic *et al.*, 2002). A recent fMRI study compared activations produced by pleasant touch, painful touch and neutral touch (Rolls *et al.*, 2003). The regions of the OFC were activated more by pleasant touch and pin prick compared to neutral touch. Interestingly the same area is known to be involved in depression (Drevets, 2000a, b), which is one of the main risk factors for developing certain types of chronic pain such as Fibromyalgia or Chronic Widespread Pain (McBeth *et al.*, 2001). These observations suggest that neural circuits involving the OFC could be implicated in the processing of emotional aspects of pain. Activity in these circuits may be greater in certain clinical pain states or when attentional resources are diverted to the affective aspect of pain.

Our demonstration of greater amygdala responses during attention to unpleasantness is supported by recent functional imaging studies in humans demonstrating amygdala activation during nociceptive processing (Bornhøvd *et al.*, 2002). Electrophysiological, neuropharmacological and lesion studies in animals also support the role of the amygdala in nociceptive processing (Bernard *et al.*, 1992, 1996; Büchel *et al.*, 1999; Schneider *et al.*, 2001). Nociceptive neurons have been demonstrated in the lateral and central nuclei of the amygdala (Romanski *et al.*, 1993). In a rat study by Shi & Davis (1999), two parallel somatic pain pathways: (i) insula-amygdala; and (ii), posterior intralaminar nuclei of the thalamus-amygdala, have been shown to be involved in fear conditioning. Also the affective areas of the cingulate cortex (area 25, 24 and 32) in primates are a projection site of the amygdala (Vogt & Pandya, 1987; Barbas & De Olmos, 1990). This supports our results of coactivation of the perigenual cingulate and

amygdala in the 'attention to unpleasantness' condition. Its precise role in nociceptive processing is unclear but it may influence the experience of fear and anxiety associated with the pain experience.

It has been suggested that the hypothalamus might act as a common autonomic and neuroendocrine output system for OFC and amygdala responses to emotions, including fear (Bohus *et al.*, 1996). Our demonstration of increased hypothalamic activity during attention to unpleasantness is consistent with this hypothesis. Hsieh *et al.* (1996b) demonstrated hypothalamic responses during severe pain and suggested that the hypothalamus could be involved in generating the autonomic output responses to the affective components of pain. This region has also been implicated in acupuncture mediated analgesia (Hsieh *et al.*, 2001). Bester *et al.* (1999) have demonstrated projections from the parabrachial nucleus to the ventromedial nucleus of the hypothalamus, which projects densely to the dorsal periaqueductal grey (PAG). This parabrachial-hypothalamus-PAG loop is thought to have a major role in aversive behaviours. Hence it is proposed that this tract might play an important role in motivational behaviour such as defence and aggression in response to noxious stimulation (Bester *et al.*, 1999). This idea may explain why few functional imaging studies in humans show hypothalamic activations, as moderate experimental pain may not evoke an aversive or aggressive response.

We have demonstrated posterior insular activation in the 'attention to unpleasantness' condition. The anterior insula is reported to be activated more consistently than the posterior insula in previous pain studies. However activations reported in insula in some of the previous functional imaging studies extend over a broad region including both anterior and posterior insula (see review by Peyron *et al.*, 2000). Posterior insular activations have been reported in many pain studies (Casey *et al.*, 1996; Valet *et al.*, 2004; also see review by Derbyshire, 2003).

We found M1 activation in the 'attention to unpleasantness' condition but not during 'attention to localization', despite the fact that subjects did not withdraw from the stimuli at any time and button press responses were matched across conditions. In addition, rCBF in M1 positively correlated with affective ratings during the 'attention to unpleasantness' condition, suggesting a relationship between M1 activity and affective processing. The M1 is classically thought to be involved in the execution of motor responses. However, although not frequently reported, some functional imaging studies of pain have shown M1 activation even in the absence of actual movement (Casey *et al.*, 1996; Lotze *et al.*, 2001; Lu *et al.*, 2004; Peyron *et al.*, 2004). Interestingly, all of these studies involved particularly unpleasant pain sensations such as visceral distension and stimulation, cold pain, neuropathic pain and capsaicin-induced pain. This is consistent with the M1 activation seen only in the 'attention to unpleasantness' condition in our study. This indicates a possible role of M1 in an affective-motivational system that may be involved in avoidance and/or escape responses.

Primary somatosensory and inferior parietal cortices

We have been able to clearly demonstrate responses in contralateral SI that are greater during 'attention to localization' than during 'attention to unpleasantness' of the same stimulus. SI is one of the main components of the lateral pain system and the response modulation is seen in the appropriate somatotopic hand area contralateral to the side of stimulation (Fig. 2).

Some functional imaging studies have failed to show activation in SI (Jones *et al.*, 1991; Bornhøvd *et al.*, 2002) or have shown decreased responses (Apkarian *et al.*, 1992) during acute pain. However, approximately 60% of functional imaging studies have shown SI

involvement in both acute and, to a much lesser extent, in chronic pain (Derbyshire, 2000). Responses have been demonstrated in both somatotopic (Andersson *et al.*, 1997) and nonsomatotopic areas (Peyron *et al.*, 2000). Somatotopic responses have also been observed in SI during anticipation of pain (Porro *et al.*, 2002). SI activations are more commonly found in imaging studies where a painful heat stimulus is moved than those with a static stimulus (Peyron *et al.*, 2000). These observations suggest that activity in SI may be dependent on attention to location, to preparatory motor activity, or both. The possibility that activity in SI could be attention-to-location-dependent is supported by recent laser pain-evoked potential (LEP) studies. These show enhanced early LEP peaks during tasks involving spatial attention to pain. Their topographic distributions are consistent with sources in contralateral SI/SII (Valeriani *et al.*, 2000; Legrain *et al.*, 2002; Schlereth *et al.*, 2003; Bentley *et al.*, 2004). The study by our group (Bentley *et al.*, 2004) was the first LEP experiment to compare attention to localization vs. unpleasantness in the same group of subjects. We demonstrated significantly greater N300 peak amplitude during 'attention to localization' compared to 'attention to unpleasantness', with no effect on the later P450 peak generated in the cingulate cortex (Bentley *et al.*, 2003). The contralateral temporal topography of the N300 component was consistent with a source in the contralateral SI/operculo-insular region. This also supports the fact that nociceptive processing occurs relatively earlier in the lateral pain system than the medial pain system (Garcia-Larrea *et al.*, 2003). The high temporal resolution of this evoked potential study complements the better spatial resolution of the PET study and supports the role of SI in the localization of pain.

Activation of the inferior parietal cortex is consistent with its role in spatial exploration and orienting in space (Karnath, 1997). It is also consistent with the role of the parietal association cortex as part of the posterior attentional system proposed by Posner *et al.* (1990).

Conclusions

It is proposed that the perigenual ACC and aMCC and associated structures are most likely to be concerned with the processing of the affective components of pain, whereas the SI and inferior parietal cortex are involved in processing sensory-discriminative component of pain. The possibility of reducing unpleasantness whilst maintaining pain localization has some therapeutic appeal. The clear identification of the areas of the pain matrix processing these components raises some hope of improving on the currently limited repertoire of pain therapies.

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Abbreviations

ACC, anterior cingulate cortex; aMCC, anterior midcingulate cortex; BA, Brodmann's area; LEP, laser-evoked potential; MI, primary motor cortex; MRI, magnetic resonance imaging; PET, positron emission tomography; pMCC, posterior midcingulate cortex; rCBF, regional cerebral blood flow; SI, primary somatosensory cortex; SII, secondary somatosensory cortex.

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